KALDRIVON

Kaldrivon MiniSMO Quick Start

Version 0.6.1 · Windows 11 x64 · © 2026 Kaldrivon

This guide gets you from download to reading real NETCONF data in about fifteen minutes. The full User Guide and the in-app Help Center (press **F1**) cover everything in more detail.

1. Install

1. Download `Kaldrivon-MiniSMO-0.6.1-Setup.exe` and its `.sha256` file.
2. Optional: check the hash in PowerShell with `Get-FileHash .\Kaldrivon-MiniSMO-0.6.1-Setup.exe` and compare it with the `.sha256` file.
3. Run the installer, accept the defaults (`C:\Program Files\Kaldrivon\MiniSMO`) and choose whether to add a desktop shortcut.
4. Select **Finish** with *Launch Kaldrivon MiniSMO* ticked.

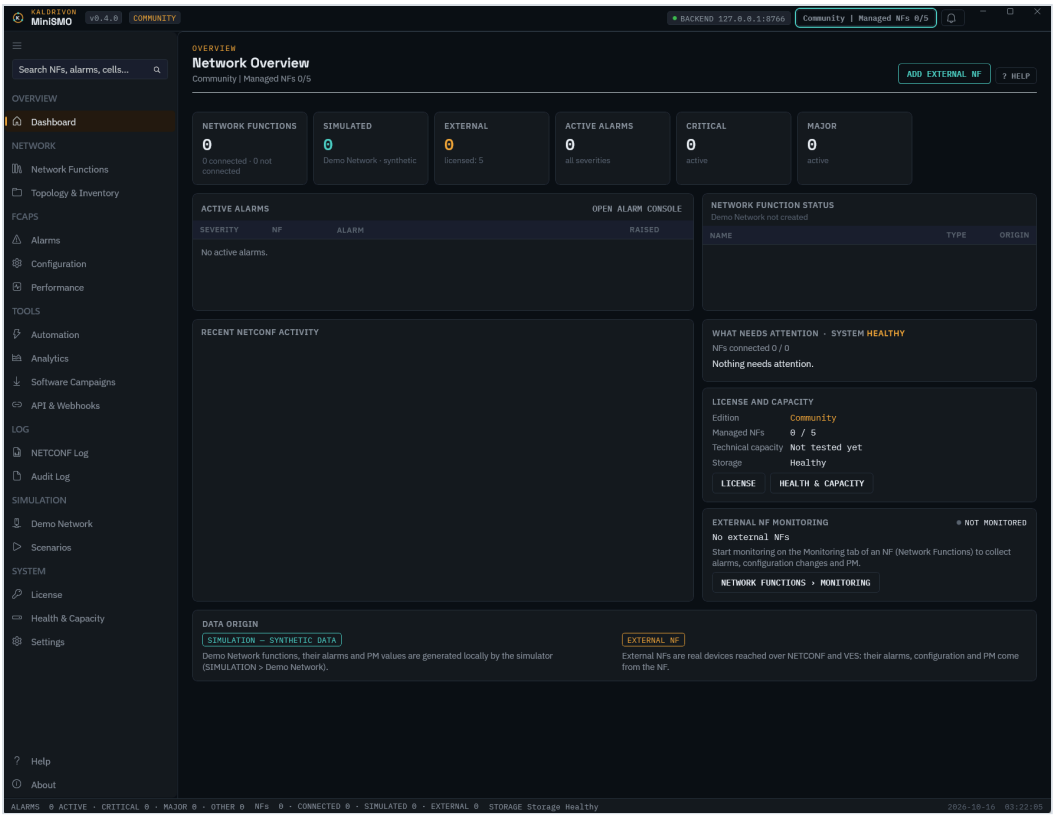
You do not need Python, .NET or anything else. Requirements: Windows 11 x64, 4 GB RAM (8 GB recommended).

2. First launch

MiniSMO starts its local backend automatically (no console window) and opens the **Dashboard**. On a new installation you see:

- `Community | Managed NFs 0/5` under the page title and in the title bar
- **ADD EXTERNAL NF** (the Demo Network is created under *SIMULATION* > *Demo Network*)

If MiniSMO reports that the backend could not start, select **Open startup troubleshooting**.



Dashboard on first launch: Community, Managed NFs 0/5

3. Create the Demo Network

Open *SIMULATION* > *Demo Network* and select **CREATE DEMO NETWORK**. MiniSMO creates 14 simulated NFs: DEMO-O-CU-CP-001, DEMO-O-CU-UP-001, DEMO-O-DU-001, DEMO-O-DU-002 and DEMO-O-RU-001 to DEMO-O-RU-010. DEMO-O-DU-001 serves DEMO-O-RU-001 to 005, DEMO-O-DU-002 serves DEMO-O-RU-006 to 010.

Simulated NFs do not count against your license: the indicator still shows `Managed NFs 0/5`.

Things to try:

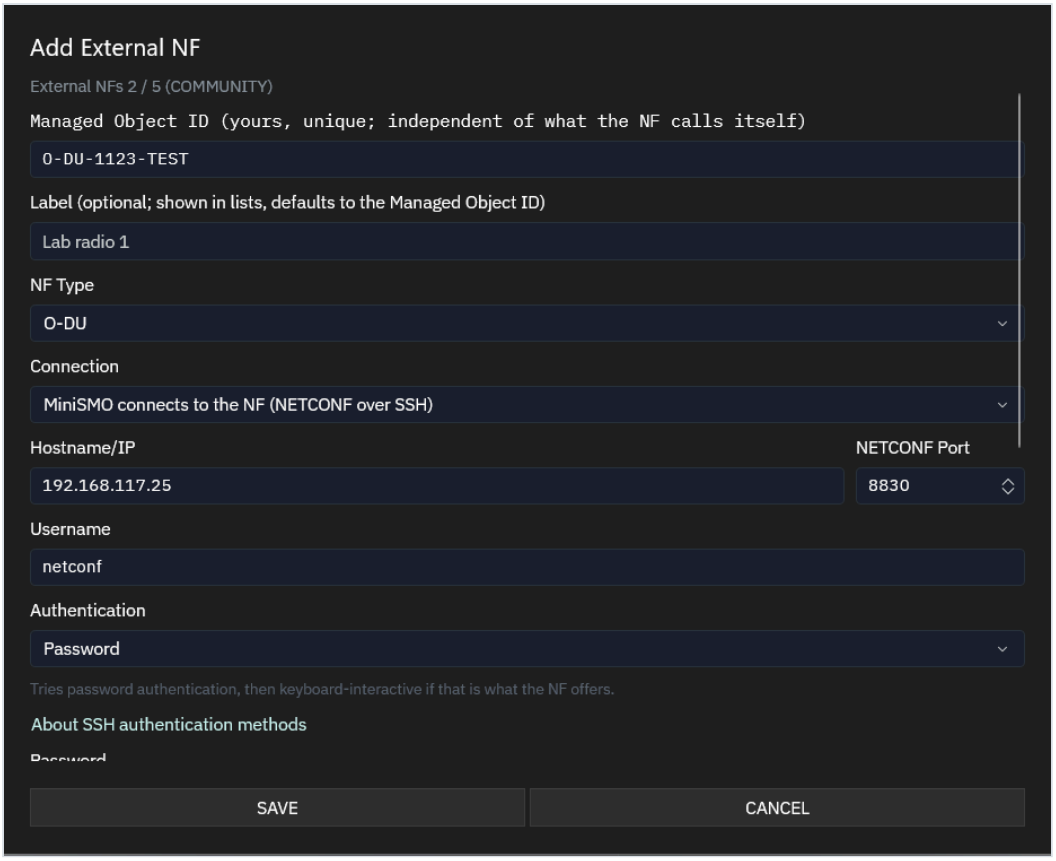
- *NETWORK* > *Topology & Inventory* to see the three layers.
- *SIMULATION* > *Scenarios*, run **CELL OUTAGE**, then look at *FCAPS* > *Alarms* and the topology. Select **STOP** when done.
- *FCAPS* > *Performance* to watch PRB, throughput and CPU move.

4. Add an external NF

You need a device (or NETCONF test server) that accepts NETCONF over SSH, usually on TCP port 830, and a username and password or SSH key for it.

1. Select **ADD EXTERNAL NF** on the Dashboard or on *NETWORK* > *Network Functions*.

2. Enter your **Managed Object ID** (for example `SITE-A/O-RU-01`; the **Label** is optional), **NF Type**, **Hostname/IP**, **NETCONF Port** (830), **Authentication**, **Username** and **Password**.
3. Select **TEST CONNECTION** (next section). It also shows what the NF calls itself (**NATIVE IDENTITY**); MiniSMO keeps that next to your ID.
4. Select **SAVE**. The indicator shows `Managed NFs 1/5`. MiniSMO then integrates the NF: it connects, starts FM, CM and PM monitoring and points the NF's VES destination at the collector. A dialog shows each step. If the device is unreachable, the NF is saved and shows **DISCONNECTED**.



Add External NF: the form filled in for an O-DU, before SAVE

Passwords are encrypted with Windows DPAPI and never shown or logged. Automatic integration can be turned off in *SYSTEM* > *Settings* > *General*, section **Integration**.

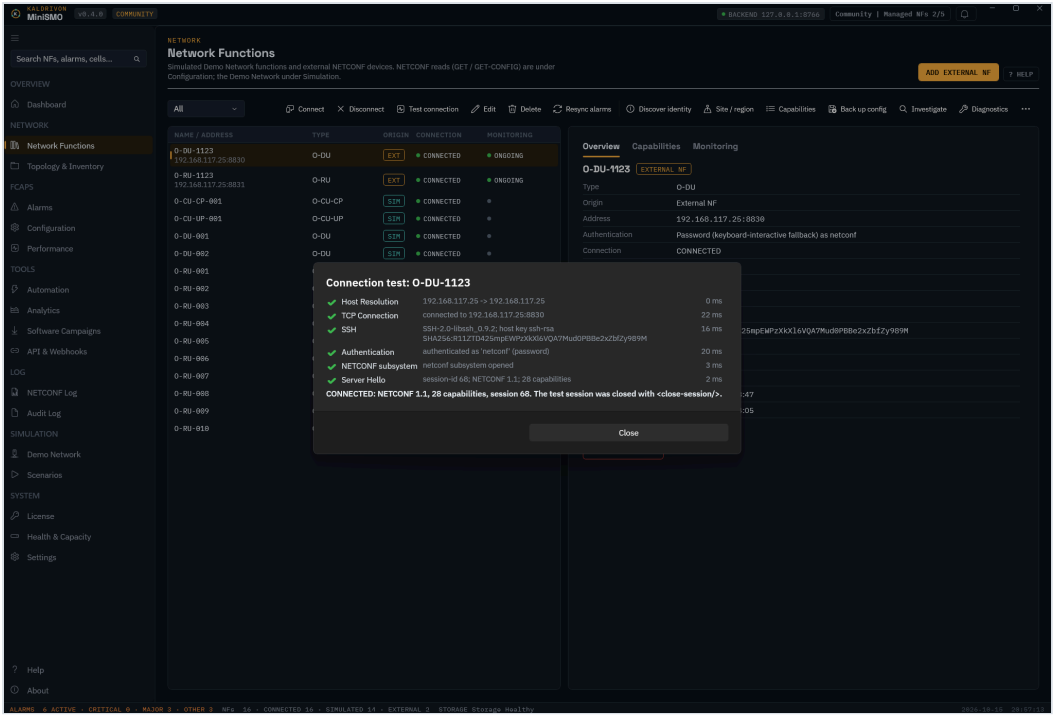
5. Test the connection

TEST CONNECTION checks six stages in order and marks each one ✓ or ✗:

Stage	If it fails, check
Host Resolution	Spelling, DNS, VPN; try the IP address
TCP Connection	Device reachable, port 830 open, firewalls
SSH	The port really is SSH

Stage	If it fails, check
Authentication	Username and password or key; the sign-in method the device allows
NETCONF subsystem	NETCONF enabled on the device for that port
Server Hello	Device NETCONF service healthy; raise the connect timeout

A full pass shows the NETCONF version (1.0 or 1.1) and the number of capabilities. **CONNECTED** is only shown after a real NETCONF hello exchange.



NETCONF connection test: all six stages passed

6. Run GET

1. Open *FCAPS* > *Configuration* and select your NF.
2. Open the **NETCONF read** tab and select **GET**.

The result shows status, message ID, session ID, timestamp, duration and the full request and reply XML (as a tree or as raw XML). The **Capabilities** tab of the NF in *NETWORK* > *Network Functions* lists every capability and YANG module the device advertised.

7. Run GET-CONFIG

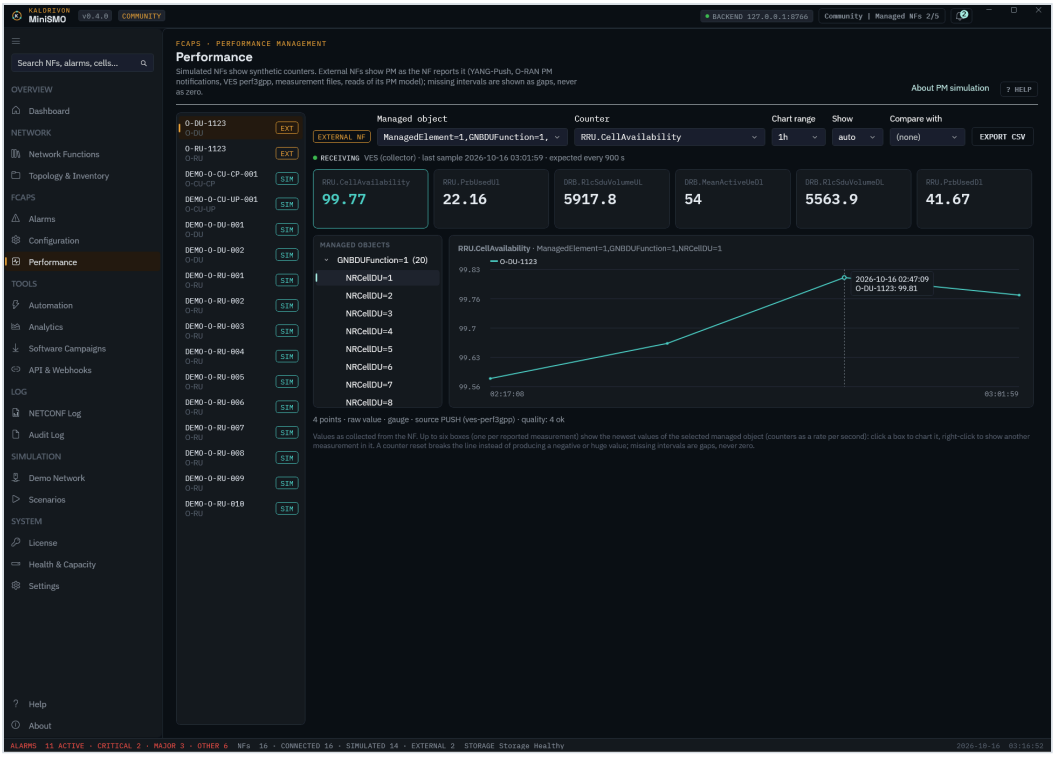
On the **NETCONF read** tab, pick a **Datastore** in the toolbar and select **GET-CONFIG**. Only datastores the device advertises can be selected (*running* always; *candidate* and *startup* only if supported).

Every NETCONF message is recorded in *LOG* > *NETCONF Log*, where you can search, filter, copy and export it.

Configuration changes on an external NF are off until you choose **ENABLE CHANGES...** for that NF on *FCAPS* > *Configuration*. Each change is reviewed and confirmed. Simulated NFs can be edited directly.

8. Monitoring

Open the NF's **Monitoring** tab in *NETWORK* > *Network Functions* to see FM, CM and PM state, the methods in use, VES events and file-based PM. Alarms appear in *FCAPS* > *Alarms*. *FCAPS* > *Performance* shows a tile per counter of the selected managed object, with the names exactly as the NF reported them, and a chart of the counter you select.



Performance: an external O-DU, counter tiles with the names the NF reported, and chart

On *FCAPS* > *Configuration* nothing is read from an external NF until you select **READ (GET-CONFIG)**.

9. License limit

All features included. **Community** (free) manages up to **5 external NFs**, **Small Deployment** up to 25, **Unlimited** has no limit. Connected and disconnected NFs both count, the same NF cannot be added twice (address and port; a DNS name and an IP address of the same device are the same NF), simulated NFs never count, and

deleting an external NF frees its slot. When you reach the limit, **ADD EXTERNAL NF** says so and nothing is added.

For more than 5 external NFs, install a Kaldrivon `.klicense` file on *SYSTEM* > *License* with **INSTALL LICENSE**. If a license expires, nothing is deleted: the 5 oldest integrations stay managed and the others are hidden (not monitored or reported) until a license is installed again. Hidden NFs are listed in *SYSTEM* > *Settings*, **Unlicensed NFs**, and offered again by **ADD EXTERNAL NF** when a slot is free.

10. Where to find Help

- **F1** on any page, or the **?** button in a page header, opens the matching help topic.
- **Help** at the bottom of the navigation pane opens the Help Center with search. Try *port 830*, *hello* or *license*.
- The **User Guide** (HTML and PDF) is installed with the application: **About** > **OPEN USER GUIDE**.
- Help works without an Internet connection.
- Support: info@kaldrivon.com · https://kaldrivon.com